

Type-II quantum spin Hall insulator without time-reversal symmetry

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Quantum spin Hall (QSH) effect is usually realized in two-dimensional materials with time-reversal symmetry, but whether it can be realized without symmetry protection, except for translational symmetry, remains unexplored. Here, we propose a type-II QSH insulator with quantized spin Hall conductivity, whose edge states with opposite chirality and polarization, distributed in different Brillouin zone regions, connect the conduction and valence bands at the boundary. Its topological invariants are characterized by the spin Chern number. Then, we demonstrate that the type-II QSH insulator can be realized in both altermagnetic materials and Luttinger compensated magnetic materials. Furthermore, based on the study of a lattice model, we find that when the spin-conserving spin-orbit coupling (SOC) effect that preserves $U(1)$ spin symmetry exists, the type-II QSH insulator phase can persist over an extensive range of SOC strengths. Even if $U(1)$ spin symmetry is broken, the type-II QSH insulator phase is robust against spin-mixing SOC. Therefore, our work not only proposes a new mechanism for realizing the QSH effect, but also enriches the types of unconventional magnetic topological phases.

quantum spin Hall effect, altermagnetism, Luttinger compensated magnetic materials, topological insulator, two-dimensional materials

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1 Introduction

Quantum spin Hall insulators (QSHIs) were theoretically proposed and then experimentally verified, which sparked widespread interest in topological research [1-4]. QSHIs

are typically realized in two-dimensional (2D) non-magnetic materials, featuring topologically helical edge states protected by time-reversal ($\mathcal{T}$) symmetry at their boundaries, as shown in Figure 1(a) [5-7]. These unique properties have significant applications in future electronic devices. An open but very desired question is whether or not we can still have a QSHI phase if $\mathcal{T}$ symmetry is broken. Although QSHIs have been proposed in antiferromagnetic materials, these materi-

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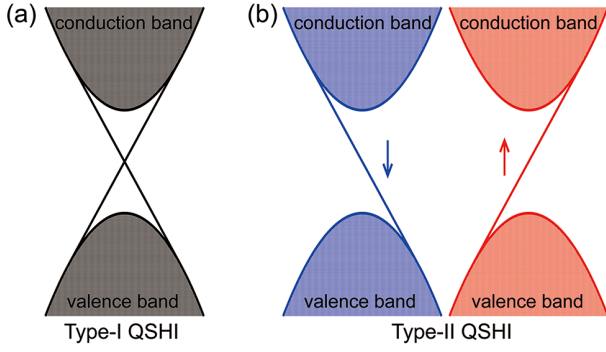


Figure 1 (Color online) (a) Conventional type-I QSHI with helical edge states protected by time-reversal symmetry in non-magnetic systems. (b) Type-II QSHI with spin-polarized edge states located at different k-points in opposite spin spaces without symmetry protection.

als all possess equivalent $\mathcal{T}$ symmetry (either the combined symmetry of spatial inversion $\mathcal{I}$ and $\mathcal{T}$, denoted as $\mathcal{IT}$, or the combined symmetry of $\mathcal{T}$ and fractional translation τ , denoted as $\mathcal{T}\tau$) [8, 9]. Therefore, it is not surprising that QSHI can also be realized in these antiferromagnetic materials. This is also the reason that altermagnetism is proposed to be distinguished from antiferromagnetism. Moreover, while a limited number of studies indicate that the QSHI can persist despite the breaking of $\mathcal{T}$ symmetry, these realizations typically rely on external inducements and only exhibit a QSHI-like behavior characterized by gapped topological edge states with mixed spin channels—distinct from the conventional QSHI, which features helical gapless topological edge states [10–13]. Thus, can the QSHI phase be realized in systems where $\mathcal{T}$ symmetry including equivalent $\mathcal{T}$ is completely broken?

Recently, altermagnetism has been proposed as a novel magnetic phase distinct from ferromagnetism and antiferromagnetism [14–16]. The spin symmetry classification and delimitation of altermagnetism [15] have provided a unifying framework to explain the previously reported ferromagnetic-like properties in compensated magnetic systems [17–21]. Due to the fact that altermagnetic materials possess both spin splitting similar to ferromagnetic materials and zero net magnetic moment like antiferromagnetic materials, altermagnetic materials exhibit many novel physical properties [14–25]. Based on the spin group symmetry, the anisotropic spin splittings in altermagnetic materials are categorized into three types: *d*-wave, *g*-wave, and *i*-wave [15], distinct from the odd-parity spin splittings (e.g., *p*-wave) enabled by non-collinear magnetic structures [26, 27]. For 2D *d*-wave altermagnetic materials, the spins of the band splitting respectively in the Γ - X direction and Γ - Y direction can be opposite to each other, which leads to new physical effects such as the crystal valley Hall effect and piezomagnetic effect, and so on [28]. As schematically shown in Figure 1(b), if a

d-wave altermagnetic material has a spin-up band inversion around the high-symmetry X point and a spin-down band inversion around the high-symmetry Y point, under spin-orbit coupling (SOC) the spin-up and spin-down bands can contribute opposite Chern numbers. When the system is projected onto a one-dimensional boundary, the high-symmetry Y and X points are projected to the $\tilde{\Gamma}$ point and $\tilde{X}$ point of the one-dimensional Brillouin zone, respectively. The opposite spin Chern numbers result in opposite chirality edge states connecting the valence and conduction bands near the $\tilde{\Gamma}$ and $\tilde{X}$ points, respectively, again as shown in Figure 1(b). It turns out that the quantum spin Hall effect (QSHE) takes place, but it does not require any symmetry protection other than translational symmetry. Here, we refer to this type of spin Hall insulator as type-II QSHI, which may be realized in altermagnetic materials or even in Luttinger compensated magnetic materials [16], in distinction from the conventional QSHI (termed as type I).

In this study, based on symmetry analysis and first-principles electronic structure calculations, we demonstrate that type-II QSHI without symmetry protection can be realized in both the 2D altermagnetic material Nb_2SeTeO and Luttinger compensated magnetic material Nb_2SeTeO under desymmetrization. Moreover, a well quantized spin Hall conductivity can also be realized in Nb_2SeTeO without and with distortion. Furthermore, through lattice model calculations and analysis, we find that when the spin-conserving SOC effect that preserves $U(1)$ spin symmetry exists, the type-II QSHI phase can persist over an extensive range of SOC strengths. Here, finite SOC generates nontrivial band topology, while $U(1)$ spin symmetry prevents mixing between the two spin sectors, allowing well-defined spin-resolved Chern numbers and quantized spin Hall conductivity. Even if the $U(1)$ spin symmetry is broken, the type-II QSHI phase remains robust against spin-mixing SOC until it transforms into an obstructed atomic insulator (OAI) phase as the SOC goes beyond a large critical value.

2 Results and discussions

2.1 Candidate materials

The monolayer Nb_2SeTeO takes a square lattice structure with space group $P4mm$ (No. 99), which is affiliated with the C_{4v} point group symmetry, as shown in Figure 2(a). It is a type of Janus material with $\mathcal{I}$ symmetry naturally broken. Our first-principles electronic structure calculations reveal that monolayer Nb_2SeTeO has a checkerboard antiferromagnetic ground state with optimized lattice constants $a = b = 4.20 \text{ \AA}$ [29–33] (The calculation results of the mag-

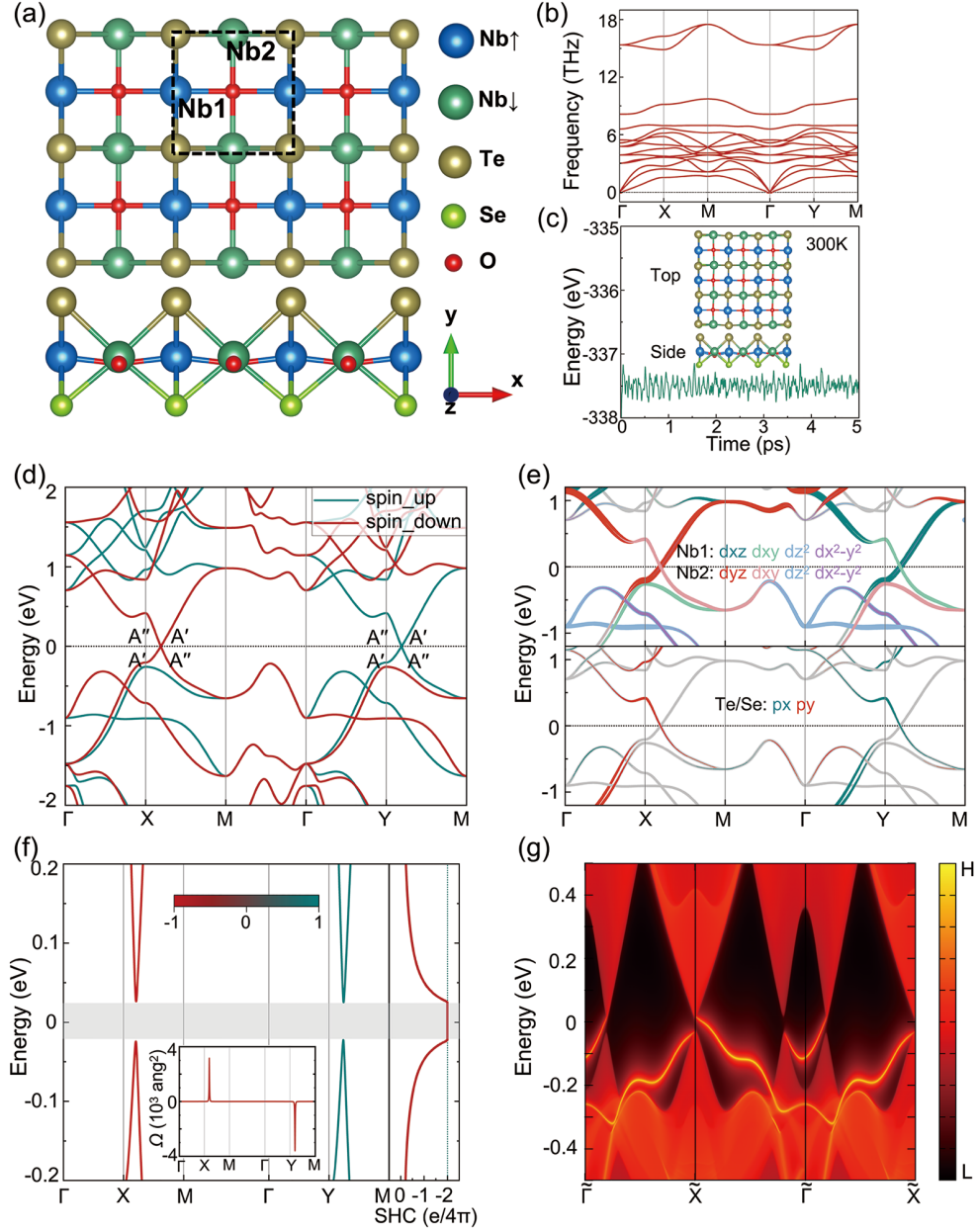


Figure 2 (Color online) (a) Top view and side view of monolayer Nb₂SeTeO. The black dashed rectangle is the primitive cell. (b) Phonon spectra. (c) The total energy evolution with respect to time in a molecular dynamics simulation. The insets are the top and side views of the final configurations for monolayer Nb₂SeTeO at 300 K after 5 ps. (d) The band structure without SOC. (e) Orbital projection of the bands without SOC. (f) The spin-resolved band structure and SHC of monolayer Nb₂SeTeO with SOC. The inset shows the Berry curvature component along the z direction of magnetization. (g) The edge states of monolayer Nb₂SeTeO cut along the (100) direction.

netic structure are presented in the Supplementary Materials). The Nb-Nb square lattice can be divided into two square sublattices, in each of which the magnetic moments of Nb atoms ($\sim 1.0\mu_B$) take their own ferromagnetic order. The dynamical and thermal stability of monolayer Nb₂SeTeO has been checked by the calculations of the phonon spectrum and *ab initio* molecular dynamics simulations [29, 30, 34]. The absence of imaginary frequency modes in the phonon spectrum, the well-maintained crystal structure, and the steady

potential energy evolution curve up to the temperature of 300 K all indicate both good dynamical and thermal stability of monolayer Nb₂SeTeO, as shown in Figure 2(b) and (c).

Since the magnetic unit cell and crystal primitive cell are the same, monolayer Nb₂SeTeO does not possess $[C_2^\perp \parallel \tau]$ spin symmetry. Moreover, due to a lack of inversion symmetry, it does not possess $[C_2^\perp \parallel I]$ spin symmetry either. However, the two Nb atoms with opposite magnetic moments can be connected by the C_4 symmetry, which implies the ex-

istence of $[C_2^+||C_4]$ spin symmetry. Therefore, monolayer Nb_2SeTeO is a d -wave altermagnetic material.

On the other hand, as shown in Figure 2(a), the Nb atoms in the sublattice with spin-up (spin-down) are connected through O (Se/Te) atoms along the x -axis and vice versa along the y -axis. Thus, there is a significant difference between the next-nearest neighbor hoppings of Nb atoms respectively with spin-up and spin-down, which can lead to a large spin splitting in monolayer Nb_2SeTeO without SOC [35]. As shown in Figure 2(d), our calculations demonstrate that a large spin splitting is indeed present along the Γ - X and Γ - Y directions, but the bands along the Γ - M direction are spin-degenerate due to $[C_2^+||M_{xy}]$ spin symmetry, which reflects the characteristics of d -wave altermagnetism. More importantly, at the Fermi level, monolayer Nb_2SeTeO has nothing but two pairs of Weyl points with opposite spins, respectively, on the X - M and Y - M axes, which are protected by $[E||M_x]$ and $[E||M_y]$ spin symmetries. Therefore, monolayer Nb_2SeTeO is a perfect bipolarized Weyl semimetal, which provides a possibility for the realization of type-II QSHE in altermagnetic materials. Meanwhile, the orbital-weight analysis (Figure 2(e)) indicates that the bands near the Weyl points are mainly contributed by the t_{2g} orbitals of the heavy element Nb, which implies that monolayer Nb_2SeTeO has a non-negligible SOC effect near the Fermi level.

When SOC is taken into account, the symmetry of monolayer Nb_2SeTeO will change from the spin point group to the magnetic point group. The specific magnetic point group symmetry depends on the easy magnetization direction of monolayer Nb_2SeTeO . Our calculations show that the easy magnetization axis is along the $[001]$ direction. Thus, the symmetry of monolayer Nb_2SeTeO includes E , $2C_{4z}\mathcal{T}$, C_{2z} , $M_x\mathcal{T}$, $M_y\mathcal{T}$, and $2M_{xy}$. Upon inclusion of SOC, the relevant symmetry changes from $[E||M_x]$ ($[E||M_y]$) to the $M_x\mathcal{T}$ ($M_y\mathcal{T}$). As a consequence, the two bands that cross at the Weyl point, which belong to different irreducible representations of the little group in the absence of SOC, are now classified by the same magnetic irreducible representation. Consequently, hybridization between these bands becomes symmetry-allowed, leading to the opening of a gap at the Weyl point due to band repulsion, as shown in Figure 2(f). In a 2D system, a massive Weyl point contributes a π Berry phase. Due to the C_{2z} symmetry, a pair of Weyl points on the X - M (Y - M) axis contributes the same Berry phase. The $C_{4z}\mathcal{T}$ (M_{xy}) symmetry causes the two pairs of Weyl points respectively on the X - M and Y - M axes to contribute opposite Berry phases (Figure 2(f)). To be specific, the pair of Weyl points on the X - M axis contributes a Chern number of 1, while the pair of Weyl points on the Y - M axis contributes a Chern number of -1 . Consequently, the spin Chern number of the system amounts to 2 [36,37]. Moreover, the spin polarization respectively on

the X - M and Y - M axes remains opposite (Figure 2(f)), that is, the bands with opposite spin polarization contribute opposite Chern numbers. Then, we also calculated the topological edge states of monolayer Nb_2SeTeO , as shown in Figure 2(g). The edge states with opposite chirality are located near the $\tilde{\Gamma}$ point and the $\tilde{X}$ point, respectively, which is consistent with Figure 1(b). Notably, since the edge states with opposite chirality do not intersect anywhere, they do not require any symmetry protection other than translational symmetry. Thus, monolayer Nb_2SeTeO is a type-II QSHI.

According to the above analysis, the opposite Chern numbers of the spin-polarized bands seem to be jointly guaranteed by the C_{2z} and $C_{4z}\mathcal{T}$ (M_{xy}) symmetries. If these symmetries are all broken, can a type-II QSHI phase still be realized? Through a shear strain that induces lattice anisotropy with $a \neq b$ and a displacement of the oxygen atom, all symmetries of monolayer Nb_2SeTeO are broken other than translational symmetry (down to the P_1 space group), but its total magnetic moment is still zero. Thus, the desymmetrized monolayer Nb_2SeTeO is a Luttinger compensated magnetic material [35,38,39]. Interestingly, when SOC is considered, the two pairs of Weyl points of desymmetrized monolayer Nb_2SeTeO still contribute opposite Chern numbers, as shown in Figure 3(a). Correspondingly, the desymmetrized monolayer Nb_2SeTeO also has two topological edge states with opposite chirality, which are located near the $\tilde{\Gamma}$ point and the $\tilde{X}$ point, respectively, as shown in Figure 3(b). Meanwhile, our calculations show that the spin Hall conductance (SHC) is well quantized to $|\sigma_{xy}^z| = 2e/4\pi$ for monolayer Nb_2SeTeO in both the pristine (Figure 2(f)) and the desymmetrized (Figure 3(a)) configurations. Therefore, the type-II QSHI phase does not require any symmetry protection other than translational symmetry and can be realized in altermagnetic materials and Luttinger compensated magnetic materials. Since the realization of the QSHE requires that the X - M and Y - M directions have opposite spin polarization. If SOC is very strong, will the QSHI still be stable? Next, we will use a lattice model to address the question.

2.2 Lattice model

In order to address the above question, we consider a tight-binding (TB) model, in which a square lattice contains two sites (labeled by sublattice index $\alpha = 1, 2$) within one unit cell. The corresponding model Hamiltonian is [35]

$$H = \sum_{d_i,j} [t^d C_{1,j}^\dagger C_{2,j+d_i} + C_{1,j}^\dagger (i\lambda^d \cdot \sigma) C_{2,j+d_i} + \text{h.c.}] + \sum_{\alpha,j} [t_\alpha^x C_{\alpha,j}^\dagger C_{\alpha,j+x} + t_\alpha^y C_{\alpha,j}^\dagger C_{\alpha,j+y} + \text{h.c.}]$$

$$+ \sum_{\alpha,j} \mathbf{m}_{\alpha} \cdot \boldsymbol{\sigma} C_{\alpha,j}^{\dagger} C_{\alpha,j}, \quad (1)$$

where $C_{\alpha,j}^{\dagger} = (C_{\alpha,j\uparrow}^{\dagger}, C_{\alpha,j\downarrow}^{\dagger})$ and $C_{\alpha,j} = (C_{\alpha,j\uparrow}, C_{\alpha,j\downarrow})$ represent electron creation and annihilation operators, respectively. As illustrated in Figure 4(a), t^d and $\mathbf{d}_{1,4} = \pm \frac{1}{2}(\mathbf{x} + \mathbf{y})$, $\mathbf{d}_{2,3} = \pm \frac{1}{2}(\mathbf{x} - \mathbf{y})$, $t_{\alpha}^{x,y}$ and $\mathbf{x} = a_1 \hat{x}$, $\mathbf{y} = a_2 \hat{y}$ represent the strength and direction of the nearest and next nearest hopping, respectively, where a_1 and a_2 are the lattice constants along the unit vectors $\hat{x}$ and $\hat{y}$. For the SOC term, we consider $\lambda^{d_1} = \lambda^{d_3}$, $\lambda^{d_2} = \lambda^{d_4}$, and $\lambda^{d_1} = -\lambda^{d_2} = \lambda$. The σ_0 and $\boldsymbol{\sigma}$ are the identity

matrix and Pauli matrix, respectively, while $\mathbf{m}_1 = -\mathbf{m}_2 = \mathbf{m}$ stands for the static checkerboard antiferromagnetic order, which couples to the electron spin as a Zeeman field. Further, $t_1^{x,y} \neq t_2^{x,y}$ and $t_1^{x,y} = t_2^{y,x}$ break the spin symmetry $[C_2^{\perp} \parallel \tau]$ but preserve spin symmetry $[C_2^{\perp} \parallel C_{4z}]$ and $[C_2^{\perp} \parallel M_{xy}]$. Since the magnetic lattice points are located at space-inversion invariant points, this lattice model also lacks spin symmetry $[C_2^{\perp} \parallel \mathcal{I}]$. Thus, this lattice model features d -wave altermagnetism.

After performing the Fourier transformation, the model Hamiltonian can be rewritten as $H = \sum_k \psi_k^{\dagger} H_0 \psi_k$ in momen-

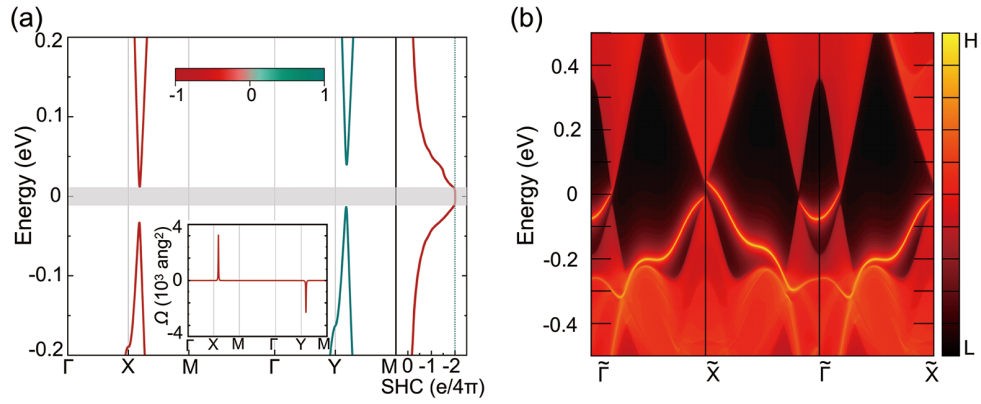


Figure 3 (Color online) For monolayer Nb₂SeTeO without any symmetry except translation: (a) The spin-resolved band structure and SHC with SOC. The inset shows the Berry curvature component along the z direction of magnetization. (b) The edge states cut along the (100) direction.

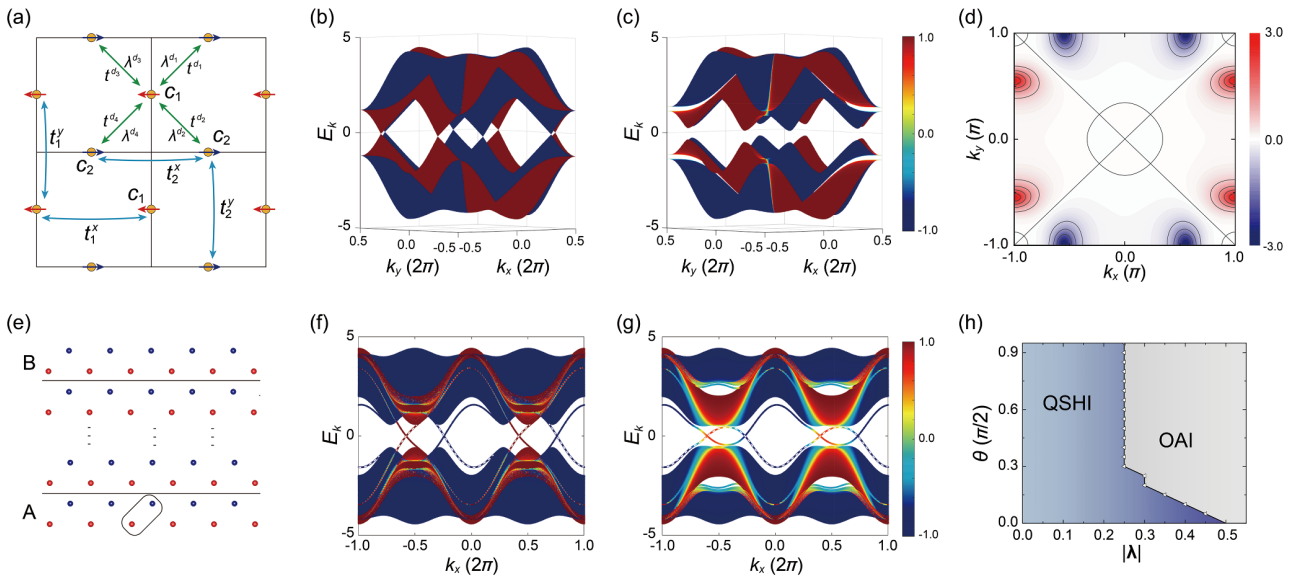


Figure 4 (Color online) The 2D lattice model with altermagnetism. (a) The schematic illustration of the hopping between lattice sites for the Hamiltonian formula (eq. (1)). The red and blue arrows represent spin-up and spin-down magnetic moments, respectively. The band structure of the lattice model without SOC (b) and with SOC (c) for $\lambda = 0.2$, $\theta = 0.2 \times \pi/2$. (d) The Berry curvature for the band structure in (c). (e) The case with open boundary conditions along the y direction while maintaining periodic boundary conditions in the x direction. (f) The corresponding edge states with the projection of S_z for (c). (g) The edge states for the obstructed atomic insulator state with $(|\lambda|, \theta) = (0.4, 0.8 \times \pi/2)$. Other parameters: $t^d = 1$, $t_1 = -t_2 = 0.8$, $|\mathbf{m}| = m_z = 1.2$. The solid and dashed lines represent the edge states of the A-cut and B-cut boundaries (The detailed results are presented in the Supplementary Materials), respectively. (h) The phase diagram of the lattice model.

tum space with basis $\psi^\dagger = (C_{1k\uparrow}^\dagger, C_{1k\downarrow}^\dagger, C_{2k\uparrow}^\dagger, C_{2k\downarrow}^\dagger)$, where H_0 reads

$$H_0 = \Gamma_k^+ \tau_0 \sigma_0 + \Gamma_k^{12} \tau_x \sigma_0 + \Gamma_k^- \tau_z \sigma_0 + \Gamma_s \tau_y \lambda \cdot \sigma + \tau_z \mathbf{m} \cdot \sigma, \quad (2)$$

in which the Pauli matrix τ_0 and τ are the identity matrix and pseudospin matrix in lattice space, respectively. The auxiliary functions $\Gamma_k^{12} = 4t^d \cos \frac{k_x}{2} \cos \frac{k_y}{2}$, $\Gamma_s = 4 \sin \frac{k_x}{2} \sin \frac{k_y}{2}$, and $\Gamma_k^\pm = (t_1 \pm t_2) \cos k_x + (t_2 \pm t_1) \cos k_y$ with $t_\alpha^x = t_\alpha$ are related to inter- and intra- sublattice hopping, respectively. The eigenvalues of eq. (2) read

$$\varepsilon_\pm^\pm(\mathbf{k}) = \Gamma^\pm \pm \sqrt{(\Gamma_k^{12})^2 + |\mathbf{m}|^2 + (\Gamma^-)^2 + \Gamma_s^2 |\lambda|^2 \pm 2A}, \quad (3)$$

with $A = \sqrt{(|\mathbf{m}| \Gamma^-)^2 + \Gamma_s^2 \|\mathbf{m} \times \lambda\|^2}$.

For $\lambda=0$, namely without SOC, the TB model (eq. (1)) is type-I bipolarized Weyl semimetal with two pairs of opposite spin polarized Weyl cones, which are protected by the spin symmetry $[E|C_{2y}]$ and $[E|C_{2x}]$, respectively, as shown in Figure 4(b) [35]. Therefore, the electronic structure of this model is consistent with that of monolayer Nb_2SeTeO without SOC. To better investigate the properties of the model under SOC, we define the angle between $\mathbf{m}$ and λ as θ , satisfying $\cos \theta = \frac{\mathbf{m} \cdot \lambda}{|\mathbf{m}| |\lambda|}$ with $\theta \in [0, \frac{\pi}{2})$.

If $\mathbf{m} \parallel \lambda$ ($\theta = 0$), all Weyl cones open gaps due to band repulsion. In this case, the SOC reduces to the $L_z S_z$ term, which commutes with S_z and therefore preserves $U(1)$ spin symmetry; consequently, S_z remains a good quantum number. Thus, the type-II QSHI phase can stably exist over an extensive range of spin-conserving SOC strengths, as shown in Figure 4(h). Once $\mathbf{m} \nparallel \lambda$ ($0 < \theta < \pi/2$), the $U(1)$ spin symmetry is explicitly broken, and spin is no longer a good quantum number (Figure 4(c)). Nevertheless, type-II QSHI is still robust against weak spin-mixing SOC, as demonstrated by our calculations of Berry curvature (Figure 4(d)) and the edge states. In the blue region of Figure 4(h), the edge states with opposite chirality connect the valence and conduction bands (Figure 4(f)), corresponding to the type-II QSHI phase. The topological invariant can still be characterized by C_s [36,37], and the spin Hall conductivity remains strictly quantized, as shown in Figure S4(b). In the gray region of Figure 4(h), with increasing SOC strength, spin-mixing becomes sufficiently strong, leading to the breakdown of quantized spin Hall conductivity, while the edge states lie completely within the band gap (Figure 4(g)), corresponding to the OAI phase [40,41]. Due to the very strong spin-mixing SOC in the gray region, the spin polarization of the bands is no longer well distinguished, and the integral of the Berry curvature over all occupied bands is zero, which accounts for the emergence of the OAI phase. On the other hand, although realistic materials always break the $U(1)$ spin symmetry, the calculations

of our lattice model show that the type-II QSHI phase can emerge within a relatively broad range of parameters (Figure 4(h)). Therefore, the type-II QSHI phase can be realized in altermagnetic and Luttinger compensated magnetic materials, for example, monolayer Nb_2SeTeO .

3 Conclusion

In summary, based on theoretical analysis, we propose the type-II QSHI, which has edge states with opposite chirality and polarization distributed in different regions of the Brillouin zone and quantized spin Hall conductivity. Subsequently, through symmetry analysis and first-principles electronic structure calculations, we demonstrate that the type-II QSHI can exist not only in altermagnetic materials but also in Luttinger compensated magnetic materials. Furthermore, based on lattice model calculations, we show that even with the breaking of $U(1)$ spin symmetry, the type-II QSHI can still exist over a broad range of parameters. Only when SOC goes beyond a large critical value, the type-II QSHI phase will transition to an OAI phase. Our work proposes a new mechanism for realizing the QSHE and also enriches the topological classes of unconventional magnetism.

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Conflict of interest The authors declare that they have no conflict of interest.

Supporting Information

The supporting information is available online at <http://phys.scichina.com> and <https://link.springer.com>. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

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